Local to Global: A Sparse Transformer-Based Small Object Detector for Remote Sensing Images

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***Abstract*— The object detection plays a crucial role in remote sensing due to the urgent demands of various applications, such as urban planning and environmental monitoring. Despite the notable progress, current methods still struggle with detecting challenging small objects. At the object level, the limited pixel representation, blurred details, and background interference of small objects impose greater demands on feature extractors. At the network level, resource bias fails to provide adequate learning signals for these objects. In this article, we propose a sparse Transformer-based detector (STDet) to tackle these challenges. Specifically, we design a local-to-global Transformer network (LGFormer) to explore essential feature representations. The local Transformer block (LTB) establishes correlations between tokens and their surrounding data, while the global Transformer block (GTB) captures long-distance dependencies related to the objects. Meanwhile, we introduce a scale-balanced label assignment (SBLA) strategy that considers more samples to small objects. SBLA dynamically shifts the learning focus to easily overlooked objects and alleviates the issue of sample imbalance. Extensive experiments on three large-scale remote sensing datasets demonstrate the effectiveness of STDet and its superiority in small object detection.**

***Index Terms*— Remote sensing object detection, small object, Transformer, label assignment.**

I. Introduction

As A fundamental and crucial image interpretation task, the remote sensing object detection technology effectively monitors and analyzes natural and artificial surface features, such as forests, rivers, buildings, and roads [1]. Advanced techniques driven by deep learning have been widely applied in agricultural monitoring, urban planning, disaster prediction, and other fields [2], [3]. However, detecting small objects continues to be a significant challenge [4]. Due to the extensive coverage of large scenes, small objects are widely distributed in remote sensing images. These objects usually occupy only a few pixels, and the presence of extensive background and irrelevant objects further impedes detection performance [5]. Detecting small objects is of considerable practical importance and remains a key challenge [6].

In the MS COCO benchmark [7], objects with a resolution smaller than 32×32 pixels are identified as small objects. Relative to the entire image, these small objects occupy only a minimal portion. In the imaging stage, the resolution of sensors directly determines the level of detail that can be captured in the remote sensing images. Small objects generally require higher spatial resolution to capture their features clearly. Meanwhile, the spectral response of small objects is usually weak. Limited by the imaging bands, these objects may exhibit spectral features similar to those of the background. Whether in the visible or infrared bands, the lower spectral reflectivity of these objects constrains their ability to be distinguished from the background in brightness and color. As a result, small objects often appear blurred in remote sensing images and are easily obscured by background noise. These factors significantly suppress the visibility of small objects, making it difficult for detectors to extract valuable discriminative features [8]. As shown in Fig. 1(b), the traditional convolutional neural network (CNN)-based ResNet [9] cannot effectively extract features from small objects. In addition, we found that compared with large objects, small objects are more difficult to generate balanced positive samples, as shown in Fig. 2(b). Insufficient samples fail to produce contributions that are significant enough for the network to prioritize [5]. This also prevents the detector from effectively considering small objects during the training stage.

In recent years, the transformer network has gained significant prominence, bringing revolutionary changes to natural language processing [11]. With the ability to capture long-range dependencies and explore global features, the transformer structure demonstrates stronger feature extraction capabilities. Modifications [12], [13], [14] have been made to the Transformer to better suit remote sensing objects. However, although the transformer has advantages in feature representation, its performance in extracting small objects is still unsatisfactory. The main reason is the uneven distribution of attention, where the prominent features of large objects

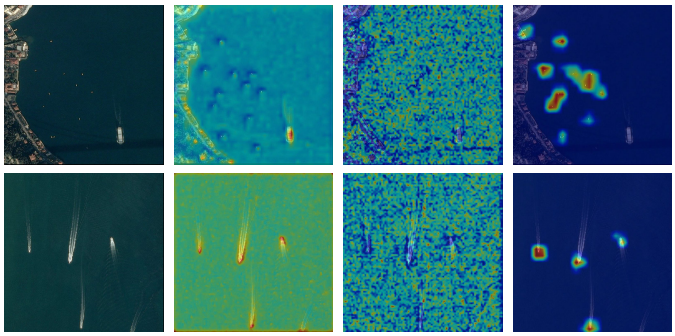
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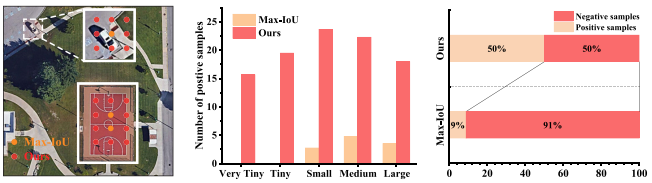
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(a) (b) (c) (d)

Fig. 1. Comparison analysis of visual feature maps from different backbones. (a) Input remote sensing images. (b) Feature heatmaps extracted by ResNet [9]. (c) Feature heatmaps extracted by Swin Transformer [10]. (d) Feature heatmaps extracted by ours.



(a) (b) (c)

Fig. 2. Comparison between our proposed SBLA and Max-IoU [17] in various aspects of label assignment. (a) Sample locations. (b) Average number of positive samples for objects of various scales. (c) Positive-to-negative sample ratio.

are more likely to attract focus in the global context. This fully global processing approach leads to insufficient attention allocation to small objects [15]. In addition, local representation is equally crucial for small objects [16].

In this article, we propose a sparse Transformer-based detector (STDet) for high-precision small object detection in remote sensing images. STDet utilizes a transformer network called local-to-global Transformer network (LGFormer) as the backbone to fully extract object features. LGFormer explores a small subset of the most relevant data related to the object to construct a global representation of overall features. This is integrated with the local features emphasized by window attention, achieving a learning strategy that progresses from local to global. Overall, LGFormer eliminates global computation, employing sparse attention to effectively reduce computational costs. To provide adequate positive samples for small objects, we propose a scale-balanced label assignment (SBLA) strategy. The SBLA strategy abandons the ideology of intersection over union (IoU) thresholds. Instead, based on the uniform assignment, it determines the number of samples each object should retain by considering the object scale and the confidence of the network predictions.

Extensive experiments are conducted on three large-scale remote sensing datasets: DIOR [18], AI-TOD [19], and NWPU VHR-10 [20]. The experimental results convincingly demonstrate the effectiveness of STDet, particularly for detecting small objects. The main contributions of this article are summarized as follows.

1) We propose a novel detector, STDet, designed to improve the detection accuracy of small objects in remote sensing images by enriching object feature representations and optimizing the evaluation of positive samples.
2) To tackle the challenge of extracting features from small objects, we introduce the LGFormer network. LGFormer captures both local and global data relevant to the object for attention modeling, facilitating effective extraction and integration of local and global features.
3) We propose the SBLA strategy to ensure a balanced assignment of positive samples across different scales. SBLA effectively addresses the scarcity of positive samples for small objects and the imbalance between positive and negative samples.

II. Related Works

A. Small Object Detection

Due to inherent characteristics and the complexity of the external environment, the small object detection in remote sensing images has consistently struggled to achieve satisfactory results [21]. Existing work mostly focuses on feature enhancement methods, which can be divided into two main approaches: feature re-expression and introducing contextual priors. Feature re-expression involves dynamically adjusting the distribution of extracted data through specific methods, such as saliency constraints [22], [23], attention mechanisms [24], [25], loss supervision [24], [26], and specific structural design [27]. These methods guide the network to learn subtle small object features from various perspectives, achieving effective results. Another approach enhances features for small objects by introducing additional contextual cues. Contextual priors refer to the implicit information about the environment and the surrounding area of the object [6]. The context-based method can be uniformly expressed as follows:

$$\hat{y} = \mathcal{P}(\mathcal{F}(f_{\text{feat}}, f_{\text{cont}}), \Theta)) \tag{1}$$

where f_{feat} and f_{cont} denote the extracted features and contextual information, while $\mathcal{F}$ and $\mathcal{P}$ denote the fusion operator and the predictor, respectively. Θ refers to the learnable parameters (Params) of the network. The distinguishability of objects gets enhanced through the aggregation of contextual features, particularly for small ones that lack sufficient information [28]. Furthermore, contextual priors, which carry strong semantic hints, can provide insights into the intrinsic properties. In complex environments, these priors serve as an important identification criterion for small objects [29].

However, the backbones adopted by most detectors fail to effectively extract the key features of small objects. Despite various feature enhancement methods, the insufficient exploitation of base features still poses a dilemma.

B. Transformer in Remote Sensing

The Transformer [11] is proposed as a deep learning model for processing sequential data. Its core unit, the self-attention mechanism, is capable of capturing long-range dependencies in texts, and its representation is as follows:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V. \tag{2}$$

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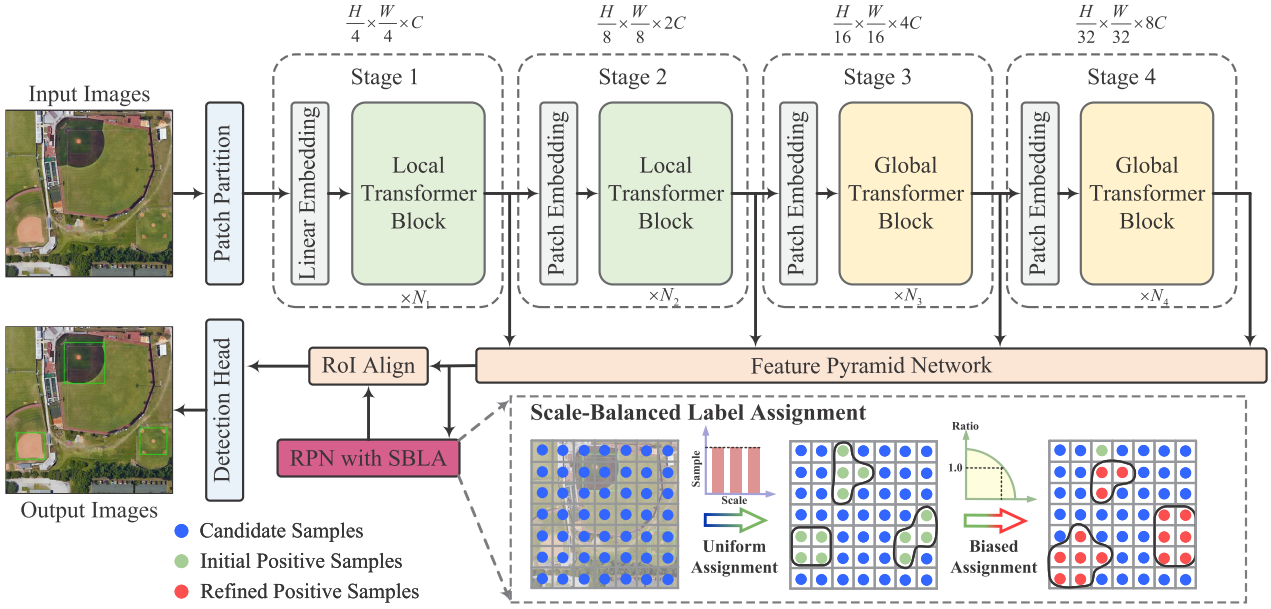


Fig. 3. Overall structure of the proposed STDet. The proposed LGFormer is responsible for feature extraction, consisting of four stages. The first two stages introduce the LTB to explore local patterns. The latter two stages incorporate the GTB to establish long-range dependencies. The proposed SBLA dynamically adjusts the number of positive samples based on object attributes.

Authors [12], [13], and [14] have shown that transformers can also play a significant role in computer vision. In remote sensing images, the extensive geographic information and complex relationships constrain traditional CNNs due to the limitations of local receptive fields. In contrast, the transformer can capture long-range dependencies and establish contextual relationships on a global scale, which is particularly beneficial for remote sensing tasks that require analyzing images from a global perspective [30]. Thus, transformers have rapidly been extended to the remote sensing community, including scene classification [31], object detection [32], image segmentation [10], and change detection [33]. However, Transformers perform poorly with small objects. This is because small objects receive less attention in global feature learning. Fully connected self-attention computations dilute the feature weights, whereas sparse attention computations are more effective for these tasks. In addition, Transformers lack local feature extraction, which is crucial for identifying the details of small objects [34].

C. Label Assignment for Small Object

Label assignment plays a critical role in small object detection, as it determines the learning targets for the network and distinguishes between foreground and background. While some detectors have specific assignment rules, most models rely on a hard threshold approach, which retains only the samples that meet certain criteria. However, this method lacks applicability for small objects due to the differences in position sensitivity between multiscale objects [35]. To address this, some work introduces flexibility in sample selection. Ge et al. [36] propose an adaptive dynamic assignment method for tiny objects, which independently determines a threshold for each ground truth (GT) based on its shape, size, and position. Kullback-Leibler divergence detector (KLDet) [37] introduces a quality score based on the Kullback–Leibler divergence localization metric and utilizes this score to effectively mine tiny object samples. Zhang et al. [38] assign more high-quality samples to tiny objects based on the proposed Gaussian distribution strategy and balance the learning of multiscale objects by adjusting the loss weights. Multistage enhancement network (MENet) [39] defines a fixed center sampling region, within which positive samples are selected based on the average overlap rate of candidate samples.

The mentioned work attempts to evaluate custom metrics to filter samples. In contrast, we assess the difficulty in identifying the GT based on the prediction results of candidate samples and use this to determine the number of samples to be selected.

III. Proposed Method

The overall structure of the proposed STDet is illustrated in Fig. 3. STDet, a high-precision two-stage object detector, is built upon Faster region-CNN (R-CNN) [17]. It consists of four main components: 1) the LGFormer backbone network; 2) the feature pyramid network (FPN) [40]; 3) the region proposal network (RPN); and 4) the detection head network. First, given an image, LGFormer extracts hierarchical features containing rich local and global information, which facilitates the subsequent network operations. Then, the features are fed into the FPN for cross-scale feature fusion. Next, the RPN generates numerous proposals based on the fused features. The SBLA dynamically divides positive samples for the network. Finally, after region of interest (RoI) align, the detection head predicts the categories and positions of the objects, outputting the final detection results. Sections III-A and III-B provide detailed descriptions of each proposed component.

A. Local-to-Global Transformer

1) Local Window Attention: Most transformer networks employ a global attention mechanism, which captures

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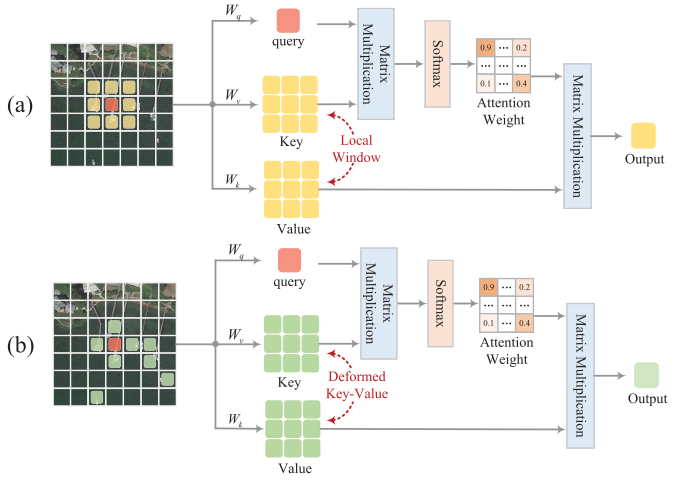


Fig. 4. Illustration of the two proposed attention mechanisms. (a) LWA. (b) GDA.

long-range dependencies and extracts global context. However, for small objects, most dependencies are local, with strong correlations typically occurring only with adjacent data. The full global computation presents the challenge of over-attention, which can suppress the saliency of small objects. This is the main reason why transformers perform poorly on small objects. In adddition, remote sensing images are much larger in size compared with natural images. In remote sensing scenarios, full-attention operations significantly increase resource consumption and inference time. To address this issue, we propose a sparse local window attention (LWA), as shown in Fig. 4(a). Inspired by convolutional operations, LWA restricts self-attention computation within a local window. Moreover, LWA is a sparse representation, where attention weight calculation is performed only between local key positions.

Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, Q, K, and V are obtained through the parallel linear projections. We specify a $\omega \times \omega$ local window centered at position (i, j) for the query. Within this window, the attention weights are computed with neighboring patches of size k. Formally, for each query in Q, there is a corresponding key–value matrix, which can be defined as follows:

$$q_{ij} = x_{ij}W_q, \quad K_{ij} = \begin{bmatrix} x_{\rho_1(i,j)}W_k \\ x_{\rho_2(i,j)}W_k \\ \vdots \\ x_{\rho_k(i,j)}W_k \end{bmatrix}, \quad V_{ij} = \begin{bmatrix} x_{\rho_1(i,j)}W_v \\ x_{\rho_2(i,j)}W_v \\ \vdots \\ x_{\rho_k(i,j)}W_v \end{bmatrix} \tag{3}$$

where $x_{\rho_k(i,j)}$ represents the kth nearest neighbor of embedding (i, j) within the local window. W_q, W_k, and W_v are the mapping matrices. K_{ij} and V_{ij} are the carefully selected local matrices for q_{ij}. They are spatially close and carry local information. Therefore, the sparse attention computation for embedding (i, j) through LWA can be defined as follows:

$$\begin{aligned} \mathrm{LWA}_k(x_{ij}) &= \mathrm{Attention}(q_{ij}, K_{ij}, V_{ij}) \\ &= \mathrm{Softmax}\left(\frac{q_{ij}K_{ij}^T}{\sqrt{d}}\right)V_{ij} \end{aligned} \tag{4}$$

where $\sqrt{d}$ is a scaling factor. LWA is repeatedly applied to all patches. Each position in the feature map can interact with local data. For patches at the edges of the feature map, we use zero padding, similar to convolutional operations, to ensure dimensional consistency. The local window controls the size of the K and V matrices, which defines the operational space of attention. If not explicitly specified, ω is set to 3. LWA can capture local patterns in features, making it suitable for small object detection with local dependencies. Meanwhile, interactions occur only within tokens in the local window, which reduces computational complexity.

2) Global Deformable Attention: To explore long-range dependencies while ensuring sparsity, we propose a global query-aware sparse attention mechanism, called global deformable attention (GDA). The schematic is shown in Fig. 4(b). We achieve this by introducing deformable convolutions [41]. Deformable convolutions set an additional convolutional layer to learn offsets $\Delta P(P_n)$ and add the learnable offsets to each convolutional position. The sampling locations of the convolutional kernel are no longer confined to fixed grid points. This can be defined as follows:

$$y(p_0) = \sum_{p_n\in\mathcal{R}} W(p_n)\cdot X(p_0 + p_n + \Delta p(p_n)) \tag{5}$$

where p_0 is the current position and $\mathcal{R}$ is the sampling region of the standard convolutional kernel $\{(-1,-1), (-1,0), \ldots, (0,0), \ldots, (+1,0), (+1,+1)\}$. Deformable convolution predicts corresponding position offsets based on position p_n, changing the original sampling location to $p_0 + p_n + \Delta P(p_n)$. The updated sampling locations maintain spatial and semantic relationships with p_0. Thus, deformed objects can be fully extracted. Inspired by the concept, we construct an offset network to search for key–value pairs related to the query.

GDA involves two key steps: deformed key–value search and sparse multihead self-attention computation. The complete process is illustrated in Fig. 5. Initially, the network predicts embedding offsets via an offset network, which are then added to the dilated positions to obtain the deformed positions. These deformed positions are semantically related to the embeddings. Subsequently, the network performs position indexing to extract sparse key–value pairs, which are fed into the sparse multihead self-attention layer for relational reasoning, generating global features.

Specifically, given an input feature $X \in \mathbb{R}^{C\times H\times W}$, we obtain the Q, K, and V matrices through linear projection. For each query in Q, we dynamically search for the most relevant k key–value pairs in K and V for attention interaction. Specifically, the offset network predicts an offset matrix $P_{\text{offset}} \in \mathbb{R}^{2k\times H\times W}$ based on the input query, where H and W are the height and width of the feature map, respectively, and $2k$ represents the coordinates of k related embeddings predicted at each pixel location. To prevent predicted offsets from becoming too large, which could interfere with the training process, we introduce dilation sampling for control. As mentioned earlier, deformable convolution predicts offsets based on position P_n. By setting a dilation rate at position P_n, the sampling regions are expanded. This allows the predicted offsets to be further minimized while

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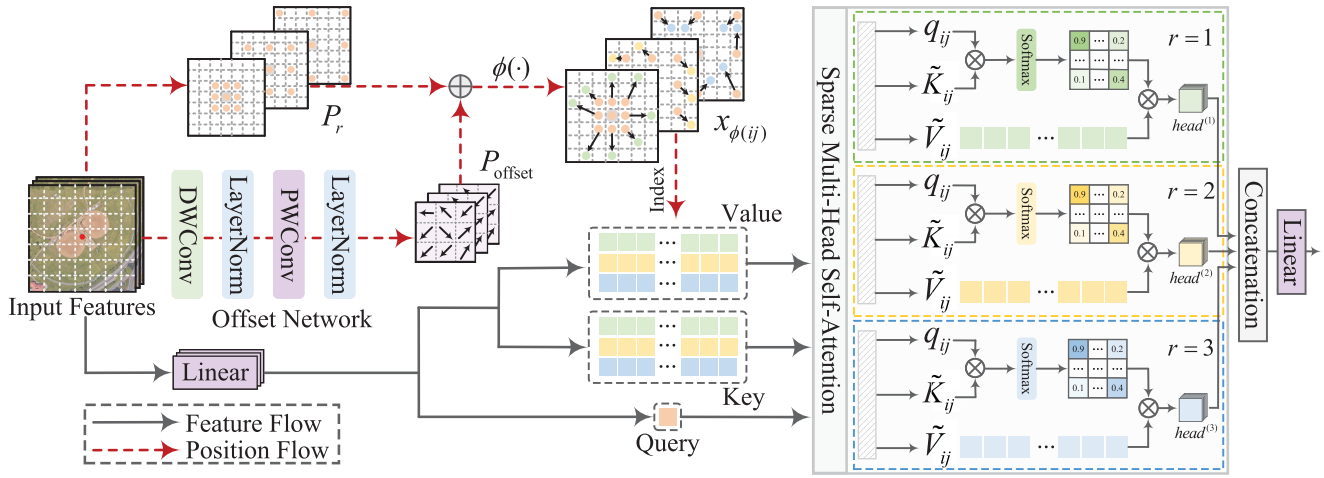


Fig. 5. Processing flow of GDA. We take three heads and dilation rates = 1–3 as an example.

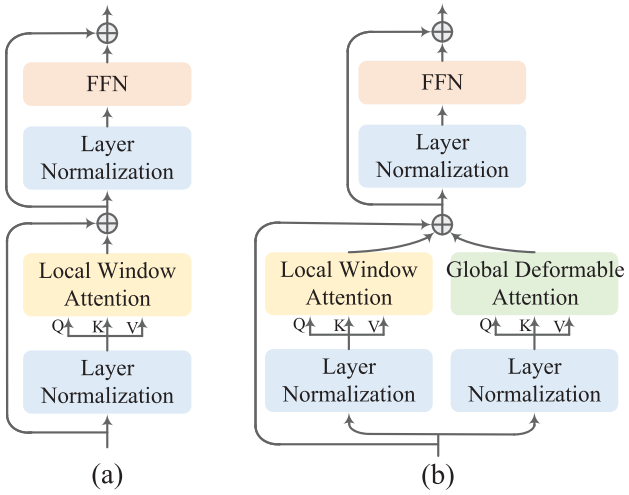


Fig. 6. Structure demonstration of the two proposed Transformer blocks. (a) LTB. (b) GTB.

still capturing global context at varying distances. At this point, the base sampling region P_r with a dilation rate of r becomes $\{(-r,-r),(-r,0),\ldots,(0,0),\ldots,(r,0),(r,r)\}$. The prediction of the relevant position can be defined as follows:

$$x_{\phi(ij,r)} = \phi(x_{ij} + P_r + P_{\text{offset}}). \tag{6}$$

The deformed positions can be any decimal values, so a rounding operation is performed to obtain the nearest grid positions. The network then indexes the K and V projection matrices based on the obtained positions to get the corresponding embeddings. Similar to LWA, GDA also represents sparse attention. The single query and key–value matrices can be defined as follows:

$$q_{ij} = x_{ij}W_q, \quad \widetilde{K}_{ij} = \begin{bmatrix} x_{\phi_1(ij,r)}W_k \\ x_{\phi_2(ij,r)}W_k \\ \vdots \\ x_{\phi_k(ij,r)}W_k \end{bmatrix}, \quad \widetilde{V}_{ij} = \begin{bmatrix} x_{\phi_1(ij,r)}W_v \\ x_{\phi_2(ij,r)}W_v \\ \vdots \\ x_{\phi_k(ij,r)}W_v \end{bmatrix} \tag{7}$$

where $x_{\varnothing_k(ij,r)}$ represents the deformed embedding. $\widetilde{K}_{ij}$ and $\widetilde{V}_{ij}$ are the global embedding matrices related to q_{ij}. Through GDA, long-distance dependencies can be established, which can be represented as follows:

$$\begin{aligned} \text{GDA}_k(x_{ij}) &= \text{Attention}(q_{ij}, \widetilde{K}_{ij}, \widetilde{V}_{ij}) \\ &= \text{Softmax}\left(\frac{q_{ij}\widetilde{K}_{ij}^T}{\sqrt{d}}\right)\widetilde{V}_{ij}. \end{aligned} \tag{8}$$

GDA can automatically locate relevant embeddings without the prior rules. Unlike deformable attention transformer (DAT) [42], our proposed GDA can establish long-range dependencies for each patch in the feature maps. Through GDA, the network can find distant data associated with small objects, thereby enhancing the global context.

3) Local Transformer Block: We introduce LWA into the transformer encoder, naming it the LTB, as illustrated in Fig. 6(a). A standard transformer encoder employs multiple self-attention heads to process data in parallel, which enhances the diversity of information extraction and improves computational efficiency. Therefore, we follow this design. Specifically, given the input x, the processing of LTB can be formulated as follows:

$$\begin{aligned} \text{head}^{(n)}(x) &= \text{LWA}(q^{(n)}, K^{(n)}, V^{(n)}) \\ \text{MultiHead}(x) &= \text{Concat}(\text{head}^{(1)}, \ldots, \text{head}^{(N)}) \\ z_{\text{attn}} &= \text{MLP}(\text{MultiHead}(x)) + x \\ z &= \text{FFN}(\text{LN}(z_{\text{attn}})) + z_{\text{attn}} \end{aligned} \tag{9}$$

where LN denotes the layer normalization layer and MLP denotes the linear projection. The FFN is a feed-forward neural network consisting of two linear layers and a Gaussian error linear unit (GELU) activation layer, with dropout introduced.

4) Global Transformer Block: To capture long-range context while maintaining local patterns, we further propose the global Transformer block (GTB), as illustrated in Fig. 6(b). We extend the LTB by introducing a GDA branch to explore global features. By adjusting the dilation rate, we can flexibly control the capture of global context across varying distances, while mitigating the uncertainty caused by excessive offsets. This approach ensures stability in learning while enhancing the completeness of global information. Given an input feature x,

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the dual-branch multihead attention is formulated as follows:

$$\begin{aligned}\text{head}_1^{(n)}(x) &= \text{LWA}(q^{(n)}, K^{(n)}, V^{(n)})\\ \text{head}_2^{(n)}(x) &= \text{GDA}(q^{(n)}, \tilde{K}^{(n)}, \tilde{V}^{(n)}, r)\\ \text{MultiHead}_1(x) &= \text{Concat}\left(\text{head}_1^{(1)}, \ldots, \text{head}_1^{(N)}\right)\\ \text{MultiHead}_2(x) &= \text{Concat}\left(\text{head}_2^{(1)}, \ldots, \text{head}_2^{(N)}\right).\end{aligned} \tag{10}$$

By adjusting the dilation rate r, the multihead structure can capture a variety of long-range contexts. Before the FFN, the two attention streams are fused through addition, integrating local and global contexts. This can be defined as follows:

$$\begin{aligned}z_{\text{attn}} &= \text{MLP}(\text{MultiHead}_1(x)) + \text{MLP}(\text{MultiHead}_2(x)) + x\\ z &= \text{FFN}(\text{LN}(z_{\text{attn}})) + z_{\text{attn}}.\end{aligned} \tag{11}$$

5) Overall Structure: The overall structure of LGFormer is depicted in Fig. 3. LGFormer successively adopts 3×3 convolutions with 2×2 strides to partition the input. The output spatial size is reduced to 1/4 of the input image. LGFormer consists of four stages, each beginning with a patch embedding layer that provides hierarchical features. Similar to a pyramid structure, as the network deepens, the patch embedding halves the spatial dimensions of the features while doubling the number of channels. This is achieved using simple 3×3 convolutions with 2×2 strides. The hierarchical features have resolutions of $(w/4) \times (h/4)$, $(w/8) \times (h/8)$, $(w/16) \times (h/16)$, and $(w/32) \times (h/32)$, respectively. In CNN-based backbones, shallow layers extract local features, while deeper layers explore global information with a larger receptive field. Following this local-to-global design, LGFormer employs LTB in the first two stages to extract local features and incorporates GTB in the final two stages to capture global features. In both LTB and GTB, attention calculations are performed in a sparse manner.

B. Scale-Balanced Label Assignment

In most detectors, such as Faster R-CNN, hard-threshold label assignment is employed. This fixed approach lacks flexibility and struggles to robustly handle biases in the training data. When dealing with small objects, it fails to ensure sufficient positive samples supporting networks during the training stage. To address this issue, we propose a quantitative SBLA strategy. SBLA adopts a uniform distribution approach, combining the scale of the objects and the network's confidence to dynamically determine the positive sample set for each GT. In addition, SBLA gradually adjusts the sample size according to the training process to improve model robustness. Algorithm 1 details the overall process of SBLA.

For each GT object $g_j \in \mathcal{G}$, an initial sample size is calculated based on the principle of average supply. It can be expressed as follows:

$$N_{\text{ave}} = \text{Int}[\![N_{\text{total}}/G]\!] \tag{12}$$

where N_{ave} is the average size of samples that should be allocated to each GT, G represents the number of GTs in a single image, $\text{Int}[\![\cdot]\!]$ represents the ceiling function, and N_{total} represents the specified total number of samples for a single image.

Algorithm 1 Algorithm of SBLA Strategy

Input:
$\mathcal{G}$ is the set of GTs
$\mathcal{A}$ is the set of preset anchors
t_{th} is the current training epoch
$\mathcal{N}$ is the specified number of positive samples
k is a hyperparameter with a default value of 9
Output:
$\mathcal{P}$ is the set of positive samples
$\mathcal{N}$ is the set of negative samples

for $t = 1$ **to** max_epoch **do**
  Calculate the t_{th} epoch positive samples $\mathcal{N}_{total}$ by Eq. (16);
  Calculate the average samples $\mathcal{N}_{ave}$ by Eq. (12);
  for each GT $g_j \in \mathcal{G}$ **do**
    Build an empty positive sample set for GT g_j: $\mathcal{P} \leftarrow \emptyset$;
    for each anchor $b_i \in \mathcal{A}$ **do**
      Calculate the IoU between b_i and g_j;
      Calculate the quality score quality b_{ij} by Eq. (13);
    end for
    Build an empty set for GT g_j: $\mathcal{S}_j \leftarrow \emptyset$;
    $\mathcal{B}_j \leftarrow$ select k anchors from $\mathcal{A}$ that have the highest quality score with g_j;
    $\mathcal{S}_j = \mathcal{S}_j \cup \mathcal{B}_j$;
    Calculate the mean quality by Eq. (14) in $\mathcal{B}_j$: $Ce(g_j) = Mean(\text{quality}(\mathcal{B}_j))$;
    Calculate the number of positive samples $\mathcal{N}_j$ to be sampled for g_j by Eq. (15);
    $\mathcal{A}_j \leftarrow$ select the top $\mathcal{N}_j$ anchors from $\mathcal{A}$ that have the high IoU with g_j;
    $\mathcal{P} = \mathcal{P} \cup \mathcal{A}_j$;
  end for
  $\mathcal{N} = \mathcal{A} - \mathcal{P}$;
end for
return $\mathcal{P}, \mathcal{N}$

However, this unified approach does not take into account the fact that there are differences in the difficulty of object prediction. Training dominated by simple objects may lose stability. Therefore, assignment rules should be directive. The sample size should be dynamically adjusted based on the identification difficulty of objects. We consider the identification difficulty from two attributes: the intrinsic characteristic and prediction complexity. Regarding intrinsic characteristics, scale variation significantly affects the network judgment. Small-scale objects are inclined to false detection. As for prediction complexity, we measure the network's certainty regarding GT based on the anchor quality. The previous method [43] employed a prediction-aware selection metric that considered the costs of classification and regression for selecting positive samples. Following this principle, we integrate both classification scores and IoU metrics to

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evaluate anchor quality. It is defined as follows:

$$\text{quality}(b_{i,j}) = \mathbb{I}[i \in S_j] \times p_{i,c_j}^{\gamma} \times \text{IoU}\left(b_i, g_j\right)^{1-\gamma} \quad (13)$$

where $\text{quality}(b_{i,j})$ represents the quality score of anchor i corresponding to GT j, c_j is the category label of GT j, p_{i,c_j} represents the predicted classification score of anchor i belonging to category c_j. b_i and g_j are the coordinates of anchor i and GT j, and γ controls the balance between the classification and regression. $\mathbb{I}[i \in S_j]$ is an indicator that specifies whether the anchor i is in the candidate set S_j of GT j.

After one-to-one matching, the quality score table is generated. This score represents the match quality between the anchor and the GT. We calculate the mean of the top k anchor quality scores corresponding to each GT to measure the prediction certainty $\text{Ce}(\cdot)$. Low certainty indicates that the object has not been fully exploited and that additional samples are needed for the network to learn

$$\text{Ce}(g_j) = \frac{\sum_{i=1}^{k} \text{quality}(b_{i,j}) \left| b_{i,j} \in \text{top}k(S_j) \right.}{k}. \quad (14)$$

Combining the intrinsic scale and prediction certainty to characterize the identification difficulty. By identification difficulty, the network is guided to focus on challenging objects, providing optimization direction for the training process. Here, we employ a simple quadratic function for this purpose. Through this function, certainty and scale are mapped into a regularized adjustment factor. This factor operates on N_{ave} to achieve dynamic sample assignment. The number of positive samples, N_j, that each GT should be assigned is defined as follows:

$$N_j = \text{Int}\left[\left[N_{\text{ave}} \times \left\{\kappa\left[\text{Ce}\left(g_j\right) \times \log_{10}^{\sqrt{w_{g_j} \times h_{g_j}}}\right]^2 + \phi\right\}\right]\right] \quad (15)$$

where κ and ϕ are the two hyperparameters that control the mapping capability. We select the top N_j anchors based on IoU scores as positive samples for GT j. Finally, a strict equal-ratio sampling of positive and negative samples is employed to ensure training balance.

To improve the discriminative capability of networks, we control the sampling of samples based on the training process. A time-dependent variable $\sigma(t_{\text{th}})$ is designed to linearly adjust the total number of samples as training progresses. The total number of positive samples N_{total} at epoch t_{th} is defined as follows:

$$\sigma(t_{\text{th}}) = \alpha + \beta \frac{T_{\max} - t_{\text{th}}}{T_{\max} - T_{\min}}$$
$$N_{\text{total}} = \text{Int}[\![N \times \sigma(t_{\text{th}})]\!] \quad (16)$$

where $T_{\max}$ and $T_{\min}$ are the maximum and minimum epochs, respectively. α and β are the two hyperparameters that regulate the range of the variable $\sigma(t_{\text{th}})$. N is the specified total number of samples for a single image. In Faster R-CNN, N is empirically set to 256. Here, we follow the same design. It is worth noting that the variable σ is linearly and inversely correlated with the progress t_{th}. As the training progresses, the total number of samples gradually reduces. This allows the network to dynamically adjust the training difficulty. Initially, the network learns a larger number of samples to cover diverse scenarios. Later, it reduces the sample size to focus on higher quality samples, enhancing the overall performance of the model. At the same time, reducing samples accelerates the training process and conserves computational resources.

IV. Experiments

A. Dataset

To comprehensively evaluate the performance advantages of STDet, three large-scale challenging datasets, namely, DIOR, AI-TOD, and NWPU VHR-10, are introduced for comparison with state-of-the-art methods.

1) DIOR: DIOR comprises 23 463 high-resolution remote sensing images with 190 288 annotated instances, designed to advance object detection technology in remote sensing imagery. It includes 20 categories: airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, golf field, ground track field, harbor, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. The images are 800 × 800 pixels in size, with a spatial resolution ranging from 0.5 to 30 m. Each instance is manually annotated using axis-aligned bounding boxes. In the experiment, the dataset is split in a 1:1 ratio into the Trainval set (11 725 images) for training and the Test set (11 738 images) for testing. For convenience, the category names are labeled as C1–C20 in the following content.

2) AI-TOD: AI-TOD is designed for detecting tiny objects in remote sensing imagery. The dataset includes 28 036 aerial images, annotated with 700 621 object instances. Each image is 800 × 800 pixels. There are eight categories: airplane, ship, person, bridge, swimming pool, windmill, storage tank, and vehicle. The average object size is only 12.8 pixels, which is significantly smaller than in other datasets. The dataset includes images of various complex scenes taken at different imaging heights. Following the MS COCO criteria, objects in the AI-TOD dataset are divided into three size ranges: [2, 8], [8, 16], and [16, 32] pixels, categorized as very tiny, tiny, and small, respectively, to facilitate the accurate evaluation of extremely small-scale objects. We utilize the designated trainval set (14 018 images) for model training and the test set (14 018 images) for performance evaluation, which is divided into a 1:1 ratio.

3) NWPU VHR-10: NWPU VHR-10 comprises 1172 high-resolution satellite images, with 2775 instances annotated by experts using horizontal bounding boxes. The dataset includes ten categories: airplane, ship, storage tank, baseball field, tennis court, basketball court, ground track field, harbor, bridge, and vehicle. Each image has a resolution of 400 × 400 pixels. The dataset is officially divided into a trainval set and a test set in a 3:1 ratio. We use the specified trainval set (879 images) for training and the test set (293 images) for model evaluation.

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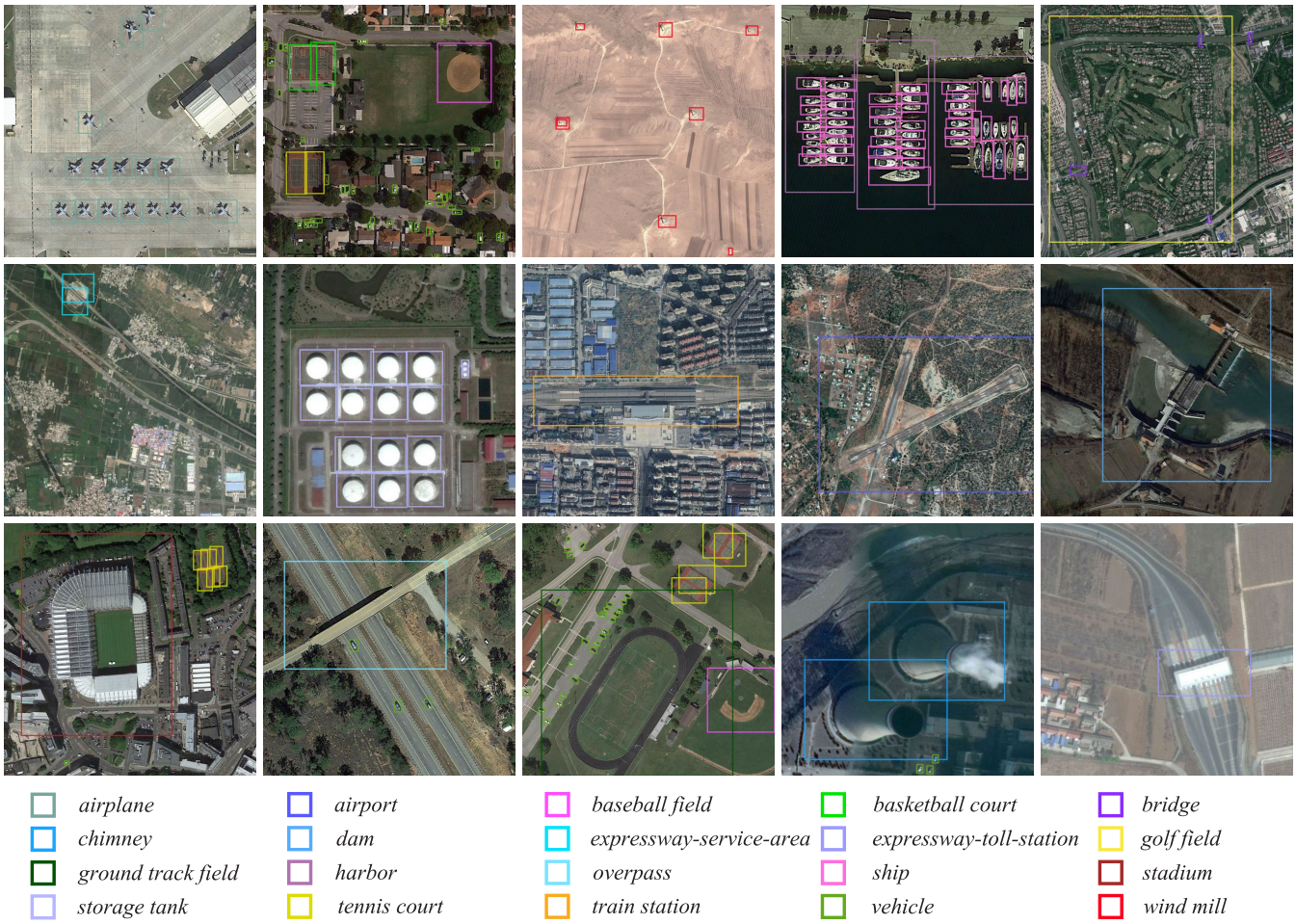


Fig. 7. Some representative predictions of STDet on the DIOR dataset.

B. Evaluation Metrics

In this article, we follow the MS COCO benchmark for evaluating the DIOR and NWPU VHR-10 datasets. The AP_{50} metric, which uses a threshold of 0.5 to determine true positives, is employed to evaluate the detection accuracy of each category. Since AP_{50} comprehensively considers both precision and recall, it accurately reflects the performance of STDet. Mean average precision (mAP) is used to evaluate the overall accuracy of the dataset, calculated as the mean of AP_{50} for each category. In ablation experiments, AP_s, AP_m, and AP_l are used to evaluate the average precision of small-scale, medium-scale, and large-scale objects, respectively. For the AI-TOD dataset, due to its unique size distribution, stricter metrics are introduced to evaluate tiny objects. Specifically, following the MS COCO benchmark, AP_{50} and AP_{75} indicate IoU thresholds of 0.5 and 0.75, respectively, for defining true positives. AP represents the average AP scores from AP_{50} to AP_{95} in 0.05 IoU intervals. To refine the evaluation of small objects (within the size range of [0, 32]), AP_s is further divided into three metrics based on object size in ablation experiments: AP_{vt}, AP_t, and AP_s, which evaluate very tiny-scale, tiny-scale, and small-scale objects, respectively.

In terms of detection efficiency, Params and frames per second (FPS) are used to evaluate the model size and detection speed.

C. Implement Details

All experiments are conducted using PyTorch and the MMDetection toolbox [58] on two RTX 3090 GPUs. For algorithm details, Faster R-CNN is adopted as the baseline. LGFormer is pretrained on the ImageNet benchmark and fine-tuned on remote sensing datasets as the backbone network. FPN is used as the neck structure. The input size of all images is uniformly resized to 800 × 800. The batch size is set to three per GPU. All experiments are optimized using AdamW, with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-4}. Random flipping is the only data augmentation method used, and all images are normalized to regulate data distribution. For the DIOR and NWPU VHR-10 datasets, the number of epochs is set to 20 and 30, respectively, with the learning rate reduced by a factor of 10 at the 12th and 18th epochs for DIOR and at the 22nd and 28th epochs for NWPU VHR-10. For the AI-TOD dataset, the number of epochs is set to 12, with the learning rate reduced by a factor of 10 at the 8th and 11th epochs. During the inference phase, the confidence threshold is set to 0.05. After NMS, the top 3000 bounding boxes are retained for object prediction.

D. Comparing State-of-the-Art Methods

1) Results on DIOR: We compare the proposed STDet with several detection methods on the DIOR dataset,

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TABLE I

COMPARISON OF DIFFERENT METHODS ON THE DIOR DATASET. THE BEST RESULTS ARE MARKED IN BOLD

Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	mAP
FSANet [44]	42.7	79.9	68.8	84.2	36.1	74.0	57.5	76.0	59.0	76.5	74.2	51.1	55.7	60.5	49.0	40.3	83.4	59.5	35.4	82.7	62.3
RetinaNet [45]	61.5	70.3	76.0	86.5	40.0	75.6	61.6	75.8	68.2	77.4	78.1	36.1	54.3	73.0	59.1	61.8	87.4	39.3	43.3	86.2	65.5
Reppoints [46]	60.5	79.1	73.4	86.2	34.8	75.8	57.1	68.2	50.6	77.2	79.6	50.4	56.2	71.7	65.5	61.3	84.4	55.4	42.3	81.0	65.5
NWD-RKA [35]	55.8	77.4	72.3	86.8	42.5	74.6	64.2	70.1	66.0	69.7	79.5	26.1	59.0	70.4	61.7	63.1	83.6	52.6	51.4	86.8	65.7
Faster R-CNN [17]	54.9	81.4	70.1	88.5	39.3	79.3	60.6	77.8	59.8	78.2	83.1	52.9	57.1	68.4	74.5	52.6	84.6	59.7	36.0	78.9	66.9
SCRDet++ [47]	64.3	79.0	73.2	85.7	45.8	76.0	68.4	79.3	68.9	77.7	77.9	56.7	62.2	70.4	67.7	60.4	80.9	63.7	44.4	84.6	69.4
MSFC-Net [48]	85.8	76.2	74.4	90.1	44.2	78.1	55.5	60.9	59.5	76.9	73.7	49.6	57.2	89.6	69.2	76.5	86.7	51.8	55.2	84.3	70.0
BSSOCNet [49]	87.4	78.4	74.5	89.1	44.3	77.6	59.8	58.5	60.2	78.2	71.8	62.4	59.2	89.7	65.7	74.0	88.5	55.6	54.4	84.8	70.7
Cascade R-CNN [50]	59.9	83.6	70.6	88.2	48.8	78.7	66.2	83.5	71.6	82.0	82.2	55.8	63.2	72.5	73.7	63.0	86.5	62.3	44.2	84.5	71.0
CFIMNet [6]	72.5	74.1	79.0	89.8	45.6	79.1	67.7	82.3	77.6	80.7	81.6	38.4	59.5	75.9	67.4	68.6	89.9	51.8	52.7	89.6	71.2
ABNet [51]	66.8	84.0	74.9	87.7	50.3	78.2	67.8	85.9	74.2	79.7	81.2	55.4	61.6	75.1	74.0	66.7	87.0	62.2	53.6	89.1	72.8
TMAFNet [25]	92.2	77.7	75.0	**91.3**	47.1	78.6	53.6	67.1	66.2	78.5	76.3	64.9	61.4	90.5	72.2	75.4	**90.7**	62.1	55.2	83.2	72.9
AGMFNet [52]	90.9	72.8	79.3	89.7	44.7	81.4	59.3	66.0	62.7	73.8	79.2	**65.0**	61.7	**91.7**	78.6	75.8	**90.7**	60.0	58.0	83.1	73.2
KLDet [37]	67.5	81.1	75.5	88.2	47.4	78.6	65.0	82.9	72.7	79.9	80.8	57.1	61.2	91.2	74.6	75.2	87.4	56.9	56.0	90.2	73.5
TBNet [53]	64.9	86.8	76.6	89.2	50.6	80.0	74.3	86.4	74.7	82.5	83.6	56.4	64.9	79.1	77.8	58.1	88.2	69.4	44.6	87.0	73.6
DetectoRS [54]	63.6	89.9	77.4	89.2	51.6	79.0	75.0	87.8	74.5	84.1	83.7	59.4	**65.5**	74.4	78.8	60.4	87.5	69.0	45.6	86.9	74.2
SESANet [55]	**93.5**	78.2	**91.9**	82.2	46.8	**89.3**	61.4	67.2	67.5	73.3	76.7	55.8	62.0	78.8	**84.8**	**82.4**	89.8	54.4	**76.7**	79.1	74.6
CoF-Net [56]	84.0	85.3	82.6	90.0	47.1	80.7	73.3	**89.3**	74.0	84.5	83.2	57.4	62.2	82.9	77.6	68.2	89.9	68.7	49.3	85.2	75.8
SFSANet [57]	87.8	83.3	87.8	86.7	48.8	78.5	68.2	85.0	76.5	73.2	78.6	62.9	63.6	90.2	83.4	79.4	83.8	61.5	67.9	84.7	76.6
STDet (ours)	72.6	**91.0**	82.8	89.5	**53.9**	80.2	**77.6**	**89.3**	**79.8**	**85.8**	**87.1**	51.6	65.4	76.1	82.1	71.0	89.2	**75.1**	58.0	**92.2**	**77.5**

TABLE II

COMPARISON OF DIFFERENT METHODS ON THE AI-TOD DATASET. 2× INDICATES THE MODEL IS TRAINED USING A 2× LEARNING SCHEDULE (24 EPOCHS). THE BEST RESULTS ARE MARKED IN BOLD

Method	Backbone	Airplane	Ship	Person	Bridge	Swimming pool	Wind mill	Storage tank	Vehicle	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
Faster R-CNN [17]	ResNet50	22.1	19.6	4.6	5.2	7.7	0.1	19.6	13.3	11.5	27.2	7.9	0.0	8.3	24.0	32.7
RetinaNet [45]	ResNet50	17.6	23.4	4.1	8.7	8.0	3.1	17.3	11.3	11.7	30.8	7.0	3.4	8.8	17.3	31.0
Cascade R-CNN [50]	ResNet50	25.6	23.6	5.3	7.4	10.8	0.0	23.3	14.1	13.8	30.8	10.5	0.0	10.6	25.5	26.6
Reppoints [46]	ResNet50	15.5	28.8	6.4	8.0	4.4	0.5	26.9	20.1	13.8	34.8	8.4	2.9	12.8	18.7	27.0
DetectoRS [54]	ResNet50	26.1	26.5	6.2	11.4	12.0	0.1	22.9	16.0	15.1	33.1	11.5	0.1	11.3	27.6	**39.4**
FSANet [44]	ResNet50	22.2	29.8	5.9	10.8	9.4	4.2	28.2	19.6	16.3	41.4	9.8	4.4	14.6	23.4	33.3
NWD-RKA [35]	ResNet50	23.4	34.2	9.5	12.6	10.9	4.1	35.1	24.2	19.2	48.3	11.2	7.2	18.9	24.7	31.6
RetinaNet w/ DLA [36]	ResNet50	11.9	49.9	5.3	17.9	14.7	6.4	27.7	21.7	19.4	50.2	11.7	8.1	19.0	23.3	22.4
KLDet [37]	ResNet50	13.6	42.3	9.3	18.7	5.2	6.7	35.7	24.9	19.6	46.4	13.7	8.4	20.6	22.7	26.4
MeNet [39]	ResNet50	22.6	39.8	9.3	16.0	10.8	6.6	35.2	22.7	20.4	50.0	12.9	8.9	21.4	23.2	31.0
TBNet [53]	ResNet50	**27.4**	43.7	10.9	18.2	15.0	7.3	36.0	25.6	23.0	54.3	15.6	9.4	24.3	28.5	32.6
STDet (ours)	LGFormer	24.9	46.6	10.5	16.4	14.0	**7.4**	37.0	25.0	22.7	54.6	15.1	11.0	24.3	24.7	32.3
$STDet_{2\times}$ (ours)	LGFormer	27.2	**56.3**	**11.5**	**19.5**	**19.1**	6.5	**38.1**	**25.9**	**25.5**	**57.5**	**18.7**	**12.4**	**26.4**	**29.0**	35.1

including general object detection methods, such as RetinaNet [45], Reppoints [46], Faster R-CNN [17], and cascade R-CNN [50] and advanced remote sensing detection methods such as FSANet [44], NWD-RKA [35], SCRDet++ [47], MSFC-Net [48], BSSOCNet [49], CFIMNet [6], ABNet [51], TMAFNet [25], AGMFNet [52], KLDet [37], TBNet [53], DetectoRS [54], SESANet [55], CoF-Net [56], and SFSANet [57]. The results are presented in Table I. STDet achieves the state-of-the-art result, reaching 77.5% mAP. It outperforms competitors with improvements ranging from 0.9% to 15.2%. Notably, compared with the latest methods AGMFNet, KLDet, TBNet, and SESANet, STDet shows improvements of 4.3%, 4.0%, 3.9%, and 2.9%, respectively. These improvements are attributed to the incorporation of LGFormer and SBLA. LGFormer has a stronger feature extraction capability compared with CNNs. It can explore subtle information and enrich feature representation, which is particularly notable in large-scale complex data. SBLA can reasonably assign samples for network learning, ensuring that overlooked instances receive attention. For detailed data, the model achieves the best performance in nine out of 20 categories, with 91.0% in airport, 53.9% in bridge, 77.6% in dam, 89.3% in expressway service area, 79.8% in expressway toll station, 85.8% in golf course, 87.1% in ground track field, 75.1% in train station, and 92.2% in windmill. We visualize some representative results in Fig. 7. It can be seen that STDet accurately predicts both large-scale structures such as stadiums and small-scale objects like vehicles. This fully demonstrates the effectiveness of our method.

2) Results on AI-TOD: To further evaluate the capability of STDet in detecting small objects, we conduct experiments on the AI-TOD dataset, which contains objects with an average scale significantly smaller than those in other datasets. We compare several representative methods to verify the advantages of our proposed method, including RetinaNet [45], Reppoints [46], Faster R-CNN [17], Cascade R-CNN [50], DetectoRS [54], FSANet [44], NWD [35], DLA [36], KLDet [37], MeNet [39], and TBNet [53]. Detailed results are reported in Table II. For overall performance, STDet achieves 22.7% AP, 54.6% AP_{50}, and 15.1% AP_{75}. The proposed method exhibits stable performance and strong robustness across various IoU thresholds, objectively confirming its effectiveness. For scale-specific performance, STDet achieves 11.0% AP_{vt} and 24.3% AP_t, outperforming the competitors. This suggests that the model is effective at handling small objects. The proposed LGFormer enhances feature representations by exploring local attention and global context. This facilitates the identification of small objects. However, the proposed method does not show a clear advantage in the AP_s and AP_m metrics. This may be because SBLA shifts the learning focus toward small objects, sacrificing some

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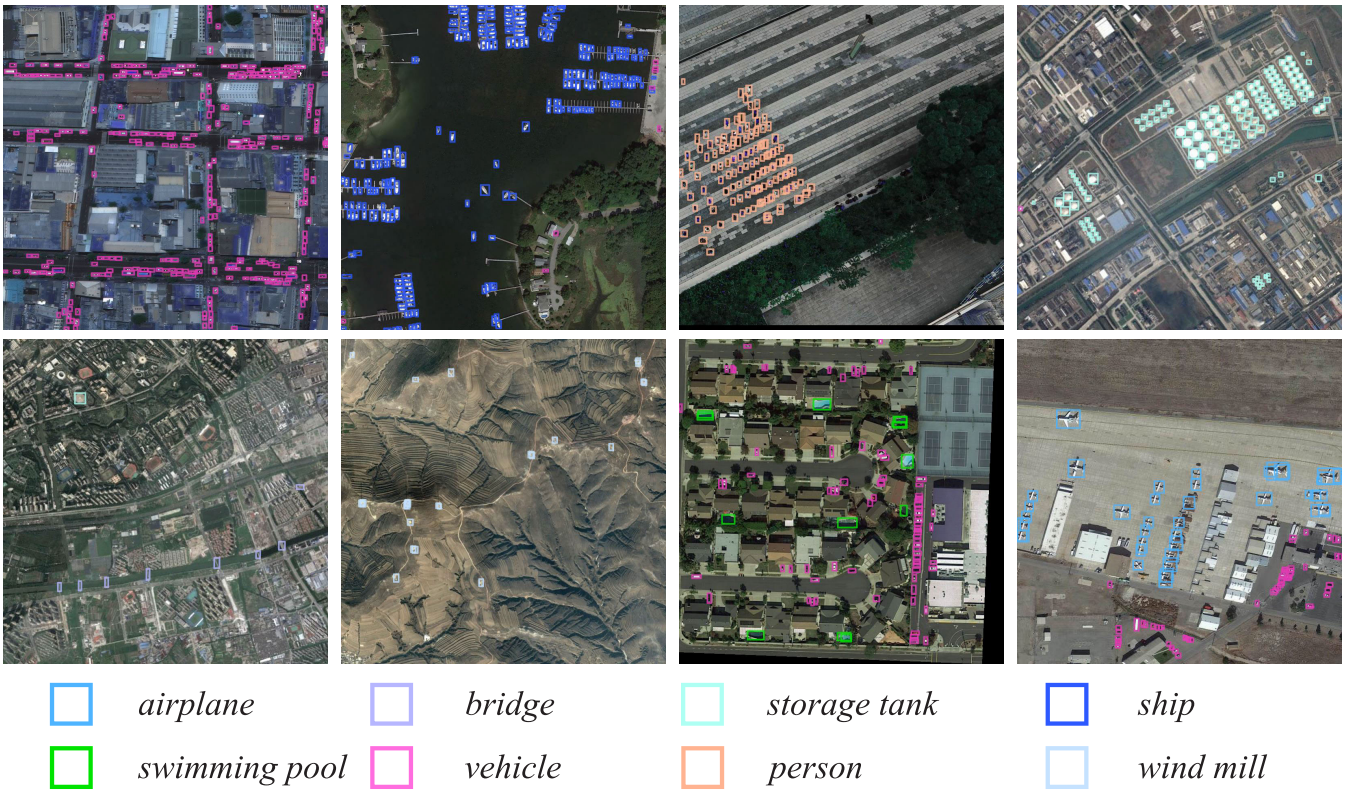


Fig. 8. Some representative predictions of STDet on the AI-TOD dataset.

TABLE III
COMPARISON OF DIFFERENT METHODS ON THE NWPU VHR-10 DATASET. THE BEST RESULTS ARE MARKED IN BOLD

Method	Backbone	Tennis Court	Bridge	Airplane	Vehicle	Storage Tank	Harbor	Baseball Diamond	Ground Track Field	Ship	Basketball Court	mAP
Faster R-CNN [17]	ResNet50	87.59	43.37	**100**	86.60	**100**	92.11	95.93	99.73	85.28	92.08	88.30
Reppoints [46]	ResNet50	90.38	63.29	**100**	86.34	96.97	87.79	95.62	97.01	91.19	86.10	89.47
FSANet [44]	ResNet50	91.84	73.88	99.92	83.42	89.24	82.81	95.93	**100**	91.09	91.07	89.92
RetinaNet [45]	ResNet50	93.47	56.43	**100**	88.63	97.14	91.72	96.23	99.93	91.70	90.54	90.58
KLDet [37]	ResNet50	94.65	70.94	99.99	93.39	96.50	82.33	95.25	99.65	92.29	91.25	91.62
DetectoRS [54]	ResNet50	93.91	86.96	**100**	88.40	92.77	92.52	96.00	**100**	91.64	**99.74**	94.19
BSSOCNet [49]	ResNet50	96.60	85.40	99.90	90.80	89.70	**97.70**	97.30	**100**	85.90	98.80	94.20
ABNet [51]	RseNet50	99.26	69.04	**100**	95.62	97.77	94.26	97.76	99.86	92.58	95.98	94.21
CoF-Net [56]	ResNet50	91.10	89.70	**100**	90.80	96.10	91.40	98.80	**100**	**95.80**	90.90	94.50
GLSANet [59]	ResNet50	**99.69**	84.83	99.90	86.76	97.12	97.51	99.40	99.46	**95.80**	86.12	94.57
SESANet [55]	DarkNet53	95.10	85.90	**100**	**96.10**	99.50	95.70	97.50	99.30	94.70	85.00	94.60
NWD-KRA [35]	ResNet50	98.85	74.56	**100**	93.84	97.64	92.90	96.95	**100**	94.58	97.03	94.64
Cascade R-CNN [50]	ResNet50	92.99	84.27	**100**	93.28	93.84	95.42	95.93	**100**	95.26	96.65	94.76
CFIMNet [6]	ResNet50	91.41	**94.95**	**100**	93.17	92.51	97.05	96.33	**100**	86.71	95.77	94.79
TMAFNet [25]	ResNet50	96.43	89.60	99.86	88.38	95.95	94.03	**99.59**	98.18	93.40	97.65	95.29
AGMF-Net [52]	GIEM	**99.69**	84.57	99.88	87.25	95.47	96.88	98.43	99.33	94.23	96.34	95.21
TBNet [53]	ResNet50	97.55	80.77	**100**	92.71	97.35	94.64	97.35	**100**	95.09	98.96	95.44
STDet (ours)	LGFormer	97.80	90.60	**100**	91.48	95.67	94.56	97.79	**100**	95.23	97.03	**96.02**

performance in others. When the training process is extended to 24 epochs, the performance of STDet improves across the board, reaching 25.5% AP, 57.5% AP_{50}, and 18.7% AP_{75}. This improvement may be due to the sample handling in SBLA, where the number of samples decreases gradually as training progresses. It allows the network to focus on high-quality samples, reducing the risk of confusion. Some representative predictions are visualized in Fig. 8. It can be observed that the size of instances for each category is quite small, especially for vehicles. Regardless of the scenario, STDet can still effectively identify these objects, which subjectively demonstrates the effectiveness of the proposed method.

3) Results on NWPU VHR-10: We also conduct experiments on the NWPU VHR-10 dataset, with the results presented in Table III. We compare several competitive methods, including Faster R-CNN [17], Reppoints [46], FSANet [44], RetinaNet [45], KLDet [37], DetectoRS [54], BSSOCNet [49], ABNet [51], CoF-Net [56], GLSANet [59], SESANet [55], NWD-KRA [35], Cascade R-CNN [50], CFIMNet [6], TMAFNet [25], AGMF-Net [52], and TBNet [53]. STDet achieves the highest performance, with a remarkable accuracy of 96.02%. It is the only method that exceeds 96%, clearly demonstrating superior performance. Our method achieves 100% AP in the airplane and ground track field categories. Moreover, STDet displays a more

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TABLE IV
ABLATION EXPERIMENTS (%) OF THE PROPOSED COMPONENTS

Baseline	LGFormer	SBLA	Backbone	AP on AI-TOD (△)	mAP on DIOR (△)	Param (M)	FLOPs (G)	FPS
✓			ResNet50	13.21 (-)	70.06 (-)	43.66	134.42	31.3
✓	✓		LGFormer	14.24 (+1.03)	75.10 (+5.04)	38.71	123.70	15.5
✓		✓	ResNet50	20.90 (+7.69)	72.10 (+2.04)	43.66	134.42	31.3
✓	✓	✓	LGFormer	22.70 (+9.49)	77.51 (+7.45)	38.71	123.70	15.5

TABLE V
COMPARE THE PERFORMANCE OF DIFFERENT BACKBONES AT THE SAME MODEL PARAM LEVEL. AP_s/PARAM REPRESENTS THE PERFORMANCE GAIN FOR SMALL OBJECTS, WITH HIGHER VALUES INDICATING BETTER PERFORMANCE

Backbone	Param (M)	FLOPs (G)	AP_s	AP_m	AP_l	mAP	$\frac{AP_s}{Param}$ ↑
ResNet [9]	23.28	52.55	25.29	57.32	87.48	70.03	1.09
FasterNet [60]	28.59	58.21	25.78	61.00	89.23	72.19	0.90
ResNext [61]	22.77	54.48	26.91	61.15	89.70	72.65	1.18
PVT [10]	**17.65**	**38.76**	25.30	57.72	90.25	71.56	1.43
Swin [32]	27.50	58.98	28.21	63.68	91.31	74.29	1.03
RMT [62]	25.42	57.83	31.46	**67.58**	91.16	**75.59**	1.24
LGFormer	21.40	45.34	**32.90**	64.99	**91.37**	75.10	**1.54**

TABLE VI
EFFECT OF SBLA COMPARED WITH THE BASELINE

Method	Optimizer	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
Baseline	SGD	11.5	27.2	7.9	0.0	8.3	24.0	32.7
	AdamW	13.2	30.0	9.9	0.0	9.3	**26.9**	**35.4**
SBLA	AdamW	**20.9**	**49.4**	**14.0**	**9.2**	**21.7**	24.1	30.8

balanced performance. It achieves over 90% accuracy across all categories, which makes it the only detector to meet this standard compared with its competitors.

E. Ablation Study

We conduct ablation experiments on the large-scale remote sensing AI-TOD and DIOR datasets to validate the effectiveness of each proposed component. For the DIOR dataset, we employ mAP (IoU = 0.5) to evaluate the overall performance. For the AI-TOD dataset, we use a more stringent metric, AP (IoU = 0.5:0.95), to assess overall accuracy. Moreover, we consider the impact of each component on efficiency using Param, floating point operations per second (FLOPs), and FPS. The experimental results are reported in Table IV. The basic ResNet50 [9] is used as the backbone, with Faster R-CNN equipped with FPN selected as the baseline. This baseline achieves 13.21% AP on the AI-TOD dataset and 70.06% mAP on the DIOR dataset.

1) Evaluation of LGFormer: In Table IV, LGFormer achieves 14.24% (↑1.03%) AP on AI-TOD and 75.10% (↑5.04%) mAP. This performance is attributed to the powerful information extraction capabilities of LGFormer, which allow it to capture multilevel and rich feature representations. These features provide discriminative information for subsequent network inference. Notably, the performance on AI-TOD is not as strong as that on DIOR, primarily due to unsuitable sample assignments. Many small objects in AI-TOD are not assigned to learning samples. Despite LGFormer extracting discriminative features, the model does not receive adequate training content. In addition, with the introduction of LGFormer, the model reduces approximately 4.95 M learnable Params and 10.72 G FLOPs.

To further evaluate the effectiveness of LGFormer, we compare it with several well-established and widely used backbones in the remote sensing community, including CNN-based models, such as ResNet50 [9] and ResNext50 [61], as well as Transformer-based models such as Swin-Tiny [10] and PVT-Small [32]. Meanwhile, we introduce the latest CNN-based FasterNet-Small [60] and Transformer-based RMT-Small [62] into the comparison. All experiments are conducted under the same level of model Param scale. The results are reported in Table V. LGFormer achieves a 75.10% mAP, slightly behind RMT. However, our model demonstrates distinct advantages in small object detection. Compared with the baseline ResNet, LGFormer shows a significant improvement of 7.61% in AP_s. Even when compared with the second-best network, RMT, LGFormer still outperforms it by 1.44%. LGFormer not only models local data relationships but also explores long-range contextual associations, which is crucial for small objects. In addition, we compare metrics such as Param and FLOPs. Except for PVT, our method shows advantages in these two aspects as well. LGFormer adopts sparse computation of related K and V pairs, which reduces computational costs while maintaining accuracy. In the last column of Table V, we compare the ratio of AP_s to Param, which we refer to as small object performance gain. LGFormer achieves the highest value among the competing networks. This indicates that, with a minimal number of Params, LGFormer delivers superior performance for small objects.

In Fig. 1, the feature responses of the three methods [9], [10] are visually compared. LGFormer effectively activates regions corresponding to small objects via local-to-global attention guidance. Compared with the CNN-based ResNet, it successfully recalls objects that are often filtered out. Compared with the Swin Transformer, the network focuses more on small objects rather than attending to global features.

In general, the main contribution of LGFormer is that it combines local and global attention to fully extract object-related information, while a sparse attention interaction alleviates the limitations of the transformer when dealing with remote sensing images.

2) Evaluation of SBLA: To validate the effectiveness of the proposed SBLA, we conduct extensive experiments on the AI-TOD and DIOR datasets. The experimental results are

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TABLE VII
GENERALIZATION ANALYSIS OF SBLA APPLIED TO VARIOUS ANCHOR-BASED DETECTORS ON THE DIOR DATASET. RESNET50 IS SELECTED AS THE BACKBONE

Methods	Param (M)	FLOPs (G)	AP_s	AP_m	AP_l	mAP
Mask R-CNN [63]	43.85	186.34	24.42	57.12	87.70	70.11
Mask R-CNN w/ SBLA	43.85	186.34	26.63	57.90	87.89	71.50
Cascade R-CNN [50]	68.98	162.24	26.12	57.48	88.40	71.04
Cascade R-CNN w/ SBLA	68.98	162.24	26.83	57.44	88.93	72.16
Libra R-CNN [64]	41.48	134.41	24.89	58.87	88.32	70.89
Libra R-CNN w/ SBLA	41.48	134.41	28.08	58.49	87.81	72.26

shown in Table IV. SBLA achieves 20.90% AP on AI-TOD and 72.10% mAP on DIOR. Compared with the baseline using Max-IoU, SBLA achieves performance gains of 7.69% and 2.04%, respectively. Compared with LGFormer, SBLA performs exceptionally well on AI-TOD but more modestly on DIOR. This discrepancy is due to the significantly smaller object sizes in AI-TOD, where the baseline struggles to assign enough positive samples under the predefined anchor and threshold rules. SBLA overcomes this by accurately selecting positive samples and addressing the underfitting issue regardless of object size. However, since object sizes in DIOR are more evenly distributed, this problem is less pronounced. In Fig. 2(b), we show the average number of positive samples determined by SBLA and the baseline across different scales. It is evident that the baseline assigns far fewer positive samples compared with our method. For very tiny and tiny-scale objects, Max-IoU often fails to designate any positive samples. This is a key factor contributing to the poor performance of small objects. In contrast, SBLA consistently provides adequate positive samples for such objects. This data further demonstrates the effectiveness of our method. In Fig. 2(c), we present the distribution of the positive-to-negative sample ratio. Max-IoU exhibits an extreme ratio, which can lead the model to predominantly predict areas as background during training. This makes it challenging for the model to distinguish between objects and backgrounds. In contrast, our method ensures balanced sampling, which enhances generalization. This is another advantage of SBLA. It is worth noting that SBLA only determines samples during the training stage. Therefore, it does not add any extra computation or Params and does not impact inference time.

We also evaluate the accuracy of SBLA across various precision metrics and object scales, as shown in Table VI. We evaluate the performance of the baseline with different optimizers. AdamW consistently outperforms SGD across the listed metrics. Thanks to its adaptive learning rate and smoother training process, AdamW provides a more stable and efficient training experience. Therefore, we select AdamW as the optimizer for performance evaluation. For overall precision metrics, SBLA achieves 20.9% AP, 49.4% AP_{50}, and 14.0% AP_{75}. Compared with the baseline, SBLA shows improvements of 7.7%, 19.4%, and 4.1%, respectively. Under various stringent conditions, SBLA provides stable performance gains, demonstrating its strong robustness. For performance across different scales, SBLA achieves 9.2% AP_{vt}, 21.7% AP_t, 24.1% AP_s, and 30.8% AP_m. Compared with the baseline, SBLA shows improvements of 9.2% and 12.4% in AP_{vt} and AP_t, respectively. This demonstrates the effectiveness of the proposed method for detecting small objects. The effective sample assignment is crucial for the detection tasks, as highlighted by the results in AP_{vt}. However, SBLA slightly trails the baseline in AP_s and AP_m. This may be because SBLA allocates more resources to challenging small objects, which shifts the learning focus and slightly compromises performance on other objects.

To further explore generalization, we integrate the proposed SBLA into various anchor-based detectors, including Mask R-CNN [63], Cascade R-CNN [50], and Libra R-CNN [64]. The experimental results are reported in Table VII. SBLA consistently improves detection performance, with gains of 1.39%, 1.12%, and 1.37%, respectively. Furthermore, we observe that the performance improvement is most significant in AP_s, which suggests that SBLA contributes notably to small object detection.

Overall, SBLA ensures an adequate supply of positive samples for each object and provides targeted preferential treatment for challenging small objects. This strategy is effectively validated through the above experiments.

F. Compatibility Analysis

To verify the compatibility between LGFormer and SBLA, we conduct ablation experiments, as shown in Table IV. When LGFormer and SBLA are combined, the performance of STDet improves significantly, reaching 22.70% AP on AI-TOD and 77.51% mAP on DIOR. Compared with the baseline, STDet achieves improvements of 9.49% and 7.45%, respectively. Notably, this performance surpasses the linear sum of the individual performances of LGFormer and SBLA, indicating that the two components are fully compatible. Features are the basis for distinguishing information, while samples are the objects of learning. LGFormer extracts distinctive features that facilitate sample identification. The high-quality samples provided by SBLA also guide the backbone to learn discriminative features. The two can synergize to unlock greater application potential.

G. Parameter Analysis

In SBLA, α and β control the sample decay rate during training. A high decay rate leads to a scarcity of samples in the later stages of training, while a low decay rate fails to achieve high-quality training. Based on this analysis, we conduct experiments and report the results in Table VIII. We find that the best performance occurs with $\alpha = 0.5$ and $\beta = 0.6$. This sensitivity experiment shows that the total amount of decayed samples maintains good performance with the specified Params. It is evident that the method remains robust to various Param values.

In (15), a simple quadratic function is chosen to map the GT attributes to a proportional factor. This choice is due to the shape of the function's descending section, which meets the adjustment requirements. Specifically, a concave function with a negative second derivative can enhance the

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TABLE VIII

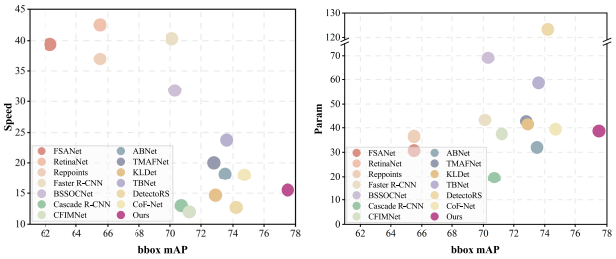
ANALYSIS OF THE PERFORMANCE IMPACT OF DIFFERENT HYPERPARAMETERS α AND β IN SBLA

α	β	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m	AP
0.4		49.2	13.3	9.2	21.5	23.4	30.7	20.5
0.5	0.6	49.4	**14.0**	9.2	**21.7**	24.1	**30.8**	**20.9**
0.6		48.8	13.5	9.0	21.4	**24.6**	30.0	20.4
	0.4	49.8	13.4	**10.5**	21.5	23.4	30.1	20.5
0.5	0.5	**50.1**	13.7	9.7	21.6	24.4	30.1	20.7
	0.7	48.2	13.6	8.9	21.2	22.6	27.9	20.2

TABLE IX

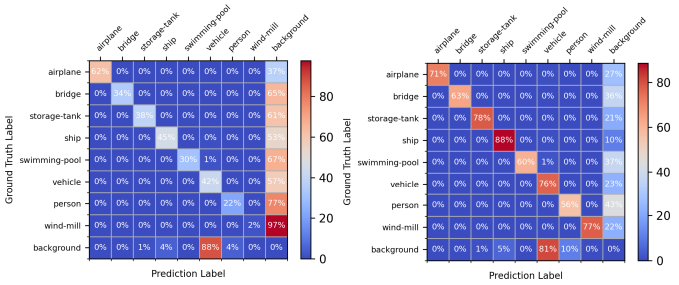
ANALYSIS OF THE PERFORMANCE IMPACT OF DIFFERENT HYPERPARAMETERS κ AND ϕ IN SBLA

κ	ϕ	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m	AP
-0.05		48.7	13.0	9.3	21.0	23.7	30.3	20.1
-0.06	1.4	**50.1**	13.7	9.7	**21.6**	**24.4**	30.1	**20.7**
-0.07		48.8	13.7	9.5	21.2	**24.4**	30.8	20.5
	1.1	49.3	**13.8**	9.7	21.5	**24.4**	**30.9**	20.6
-0.06	1.2	49.3	13.6	9.4	**21.6**	24.3	30.5	**20.7**
	1.3	49.1	13.6	9.1	21.3	22.8	30.8	20.3
	1.5	47.6	**13.8**	**10.1**	21.5	22.7	28.7	20.3



(a) (b)

Fig. 9. Comparison of detection performance (bbox mAP), inference speed, and Params of various methods on the DIOR dataset. (a) Comparison of bbox mAP versus speed. (b) Comparison of bbox mAP versus Param.



(a) (b)

Fig. 10. Comparison of the confusion matrix between our STDet and the baseline on the AI-TOD dataset. (a) Baseline. (b) Ours.

adjustment intensity as the independent variable increases. Here, κ and ϕ are the two tunable hyperparameters that control the adjustment range. We conduct Param experiments, as shown in Table IX. With $\kappa = -0.06$ and $\phi = 1.4$, the model performs excellently and achieves a more balanced performance. At this point, difficult-to-detect small weak objects receive more positive samples, while the network slightly reduces samples for other objects to shift the learning focus.

H. Discussion

1) Accuracy Versus Efficiency: In Fig. 9, we compare various detectors on the DIOR dataset in terms of accuracy (bbox mAP), inference speed, and Params. General detectors, such as Faster R-CNN [17], RetinaNet [45], and Reppoints [46] typically exhibit high efficiency. However, due to the lack of specific components for remote sensing data, their accuracy is somewhat limited. In contrast, remote sensing-specific detectors such as TBNet [53], DetectoRS [54], and CoF-Net [56] incorporate tailored domain adaptations, achieving superior accuracy at the expense of efficiency. Our proposed STDet involves frequent positional indexing, which, while not introducing additional training Params, incurs higher inference overhead. In future work, we will aim to optimize this procedure to balance accuracy and efficiency better.

2) Confusion Matrix: We visualize the confusion matrix on AI-TOD to intuitively display the model performance. In the confusion matrix, the rows represent the truth categories in the dataset, while the columns represent the predicted categories for each sample. The data on the diagonal represents the true positive rate, i.e., the prediction accuracy. The last column of the matrix shows the false negative rate for each category, reflecting the missed detection rate. The last row of the matrix indicates the proportion of negative samples that are incorrectly predicted as positive by the model. In Fig. 10, it is clear that the baseline exhibits a high missed detection rate across all categories. Except for airplane, the prediction accuracy for other categories is below 50%, meaning that more than half of the objects are misclassified. Specifically, the missed detection rates for windmill and person reach 97% and 77%, respectively. This is due to the small size of these objects, which makes it difficult for the network to extract valuable features. In contrast, the proposed STDet improves prediction accuracy for all categories. For instance, the accuracy for windmill increases from 2% to 77%. Missed detection rates for other categories also decrease substantially. This fully demonstrates the effectiveness of STDet.

3) Precision Versus Recall Curves: Precision versus recall (PR) curves illustrate the tradeoff between the model's accuracy in identifying objects, and its ability to cover all relevant instances. The shape of the curve provides an intuitive reflection of the overall performance. The area under the PR curve (AUC-PR) quantifies this performance, with a higher AUC-PR indicating a better-performing model. We visualize the PR curves for each category and the different maximum prediction counts (maxDet) on AI-TOD in Fig. 11. As observed, when maxDet = 10, STDet detects more objects across all categories compared with the baseline. This indicates that the proposed model has higher prediction confidence and stability. As maxDet increases to 100, STDet still performs better than the baseline, demonstrating its capability to detect more objects. In category-specific performance, the proposed method consistently improves detection performance across all categories. Specifically, for dense and complex categories such as person, vehicle, and ship, the baseline struggles, whereas STDet maintains a high recall rate with fewer predicted boxes. For sparse categories such as airplane and storage tank, the

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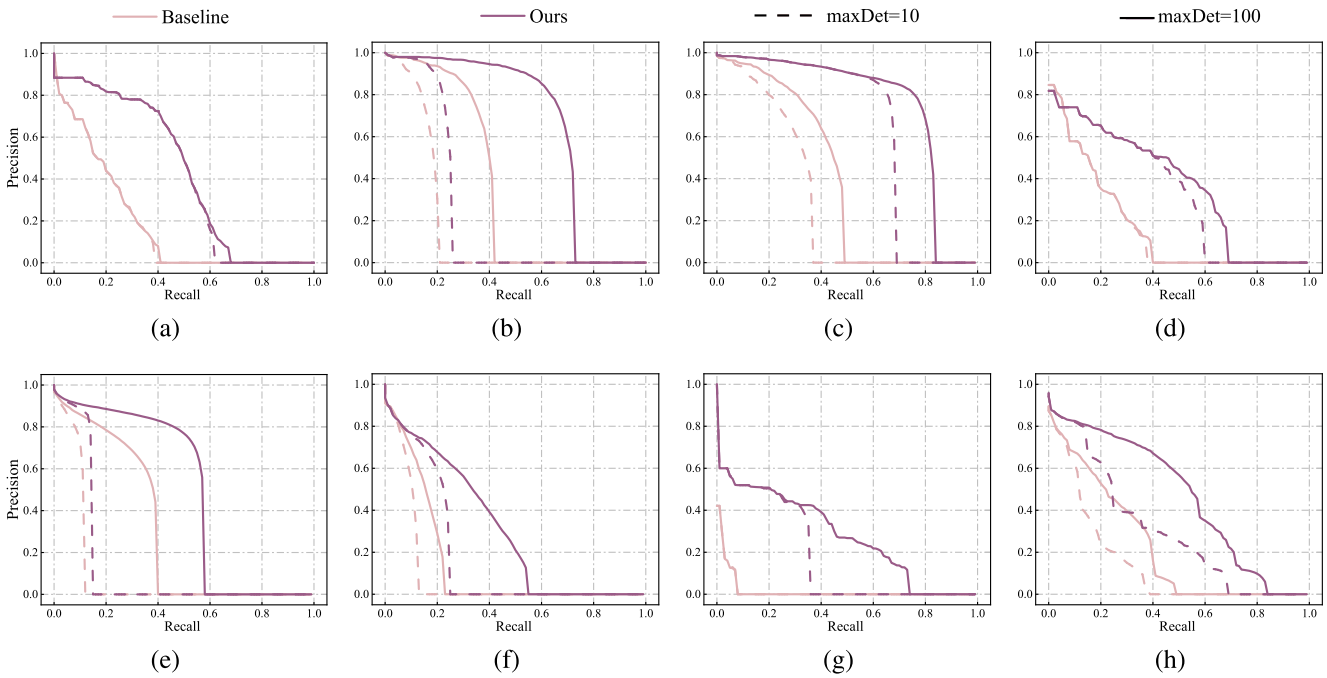


Fig. 11. PR curves for our proposed model and baseline for different categories and maximum number of predictions per image (maxDet). Best viewed in color. (a) PR curve at 0.5, airplane. (b) PR curve at 0.5, bridge. (c) PR curve at 0.5, storage tank. (d) PR curve at 0.5, ship. (e) PR curve at 0.5, swimming pool. (f) PR curve at 0.5, vehicle. (g) PR curve at 0.5, person. (h) PR curve at 0.5, windmill.

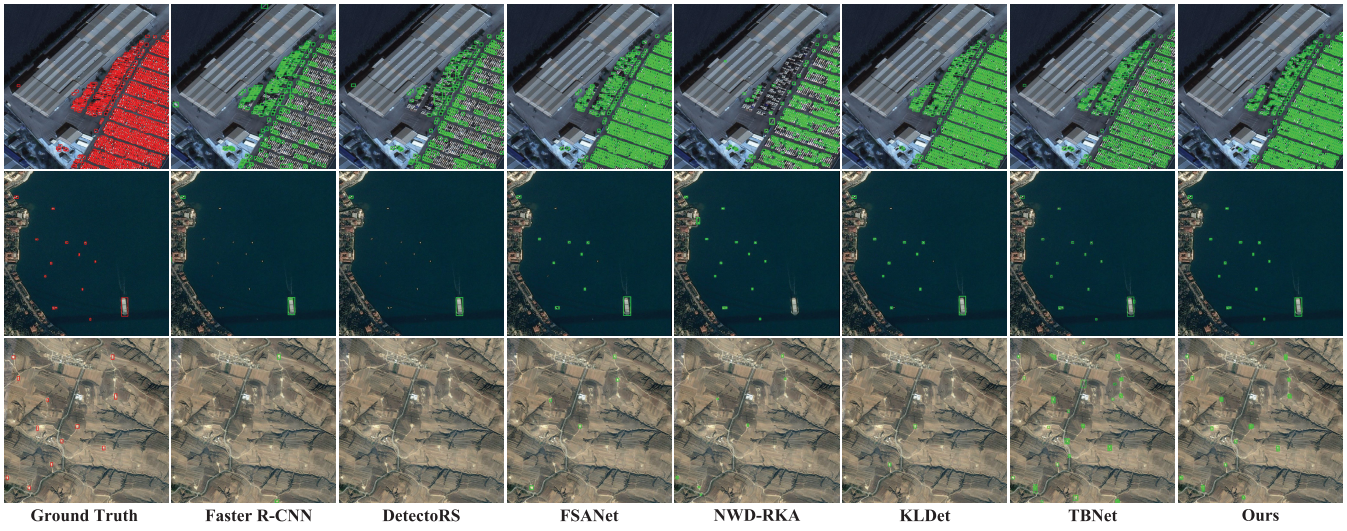


Fig. 12. Qualitative comparison results of different methods on the AI-TOD dataset.

proposed method further improves detection performance over the baseline. These results validate the advantages of the proposed method in small object detection.

4) Qualitative Comparison: We visualize the detection results of six methods across three complex scenarios (parking lot, maritime, and desert) on AI-TOD for qualitative comparison. As shown in Fig. 12, Faster R-CNN [17] and DetectoRS [54] suffer from significant miss detections in all environments. FSANet [44], NWD-RKA [35], and KLDet [37] maintain high recall rates in dense parking lot areas. However, when further processing weak ships and windmills that are highly similar to the background, these methods still tend to make false positives. TBNet [53], while correctly detecting more objects, generates a large number of false negatives when extracting similar background data. In contrast, the proposed STDet effectively localizes small-scale and visually ambiguous objects with high precision in complex scenarios. These cases reveal the superior capabilities of our proposed method.

I. Application Scenarios

To evaluate the performance of the proposed method in real-world applications, we select large-scale, challenging remote sensing images from the DOTA [65] dataset for prediction. We use the training Params of AI-TOD to execute this work. The entire image is fed into the network to assess its generalization capabilities. To evaluate the detection performance, we compare the prediction results against the baseline, with visualizations shown in Fig. 13. It is clear

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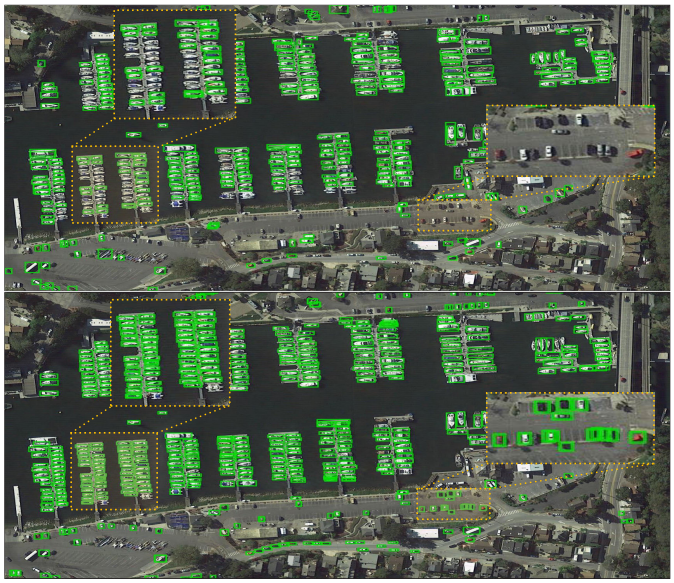


Fig. 13. Visual comparison between the proposed STDet and the baseline on large-scale remote sensing images in the DOTA dataset. The prediction results of the baseline are shown in the top image, and ours are shown in the bottom image.

that our method performs well in this task. Compared with the baseline, STDet accurately predicts more objects. We highlight some notable results for a clearer visual comparison. The proposed method successfully detects ships at sea and small vehicles on the shore—objects that the baseline largely misses. This demonstrates that the proposed method effectively handles densely distributed small objects in complex backgrounds, which are of significant value in practical decision-making. Even when applied to real large-scale data, STDet maintains high accuracy and strong generalization.

V. Conclusion

In this article, we introduce an STDet tailored for the small object detection in remote sensing images. First, we design a novel backbone network, LGFormer, which adopts a local-to-global paradigm to extract critical features progressively. Its integrated window attention mechanism facilitates sparse interactions within local data, while the proposed GDA captures long-range contextual dependencies. This network can focus attention on sparse regions related to small objects. Second, we propose an SBLA strategy that dynamically determines the sample assignment based on the object scale, identification difficulty (from classification and localization prediction), and training process. This strategy allocates more learning resources to small objects. Finally, we validate the proposed STDet through extensive experiments on three large-scale remote sensing datasets: DIOR, AI-TOD, and NWPU VHR-10. The results fully demonstrate the effectiveness and advantages of our method. In future work, we plan to build upon the remote sensing imaging context and further improve small object detection by integrating advanced techniques, such as multimodal learning and image super-resolution.

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